

Adult Odonata conservatism as an indicator of freshwater wetland condition



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ABSTRACT

There is a growing need to identify effective and efficient biological indicators for wetland assessment, and adult damselflies and dragonflies (Insecta: Odonata) possess several attributes that make them attractive for this application. We introduce a general indicator of freshwater wetland condition based on objectively estimated adult Odonata species conservatism, or sensitivity to human disturbances. We used an extensive opportunistic survey dataset from Rhode Island (USA) to empirically assign a coefficient of conservatism (CoC) to each of 135 Odonata species, based on their exclusivity to categories of degradation among 510 wetlands; the mean CoC of species observed in the adult stage was applied as an index of wetland integrity. An independent sample of 51 wetlands was also drawn from the opportunistic survey to test the performance of the index relative to human disturbance, as measured by multimetric rapid assessment and surrounding impervious surface area. The index was well predicted by both disturbance measures and showed no evidence of dependence on sampling effort, wetland size, or geomorphic class. Our findings suggest that conservatism of adult Odonata averaged across species may provide a robust indicator of freshwater wetland condition. And because adult Odonata are generally easy to identify, especially relative to larval Odonata, the index could be particularly useful for wetland assessment. Our straightforward empirical approach to CoC estimation could be applied to other existing spatially referenced Odonata datasets or to other species assemblages.

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1. Introduction

Biological indicators (or bioindicators) can provide reliable, quantitative characterizations of ecological condition, and there is a growing need to identify effective bioindicators for use in wetlands management and protection (Sifneos et al., 2010; U.S. EPA, 2002). Macroinvertebrates have long been recognized as useful bioindicators for aquatic and wetland ecosystems (Hilsenhoff, 1977; Karr and Chu, 1999; Rader et al., 2001; Wissinger, 1999), but the impracticalities of collecting, sorting, and identifying aquatic stages limit their use in rapid assessments (Cummins and Merritt, 2001; King and Richardson, 2002; Turner and Trexler, 1997). It is therefore worthwhile to evaluate taxa and life stages that are both ecologically important and logistically feasible for bioassessment. Aerial stages of aquatic macroinvertebrates are important for species dispersal and the transfer of energy across aquatic and upland systems and among trophic levels (Malmqvist, 2002; Sanzone et al., 2003), and are more sensitive than the aquatic stages to land use practices

around wetlands (Anderson and Vondracek, 1999; Raebel et al., 2012; Tangen et al., 2003).

Dragonflies and damselflies (Odonata) are prominent in many freshwater habitats and may contribute a large proportion of total invertebrate biomass and species richness (e.g., Batzer et al., 1999; Blois-Heulin et al., 1990; Rader et al., 2001; Sang and Teder, 2011; Wittwer et al., 2010). Odonates are sensitive to conditions at the breeding site and surrounding terrestrial area, can react quickly to changes in environmental quality via active dispersal, and contain a tractable number of species for practical use (Chovanec and Waringer, 2001; Oertli, 2008). Adult odonates are conspicuous over water and relatively easy to identify at the species level (Bried et al., 2012a; Oertli, 2008; Raebel et al., 2010), and may be especially well suited for broad and integrative assessments of the wetland breeding site and surrounding landscape (Bried and Ervin, 2006; Dolný et al., 2012; Foote and Hornung, 2005; Foster and Soluk, 2006; Reece and McIntyre, 2009). Adult odonates are therefore well-suited for rapid assessment methods (Fennessy et al., 2007) and addressing the increased focus on wetland quality and not just quantity in the United States (Scozzafava et al., 2011).

Odonata are already established as focal organisms for freshwater conservation (Samways, 2008) and as good indicators of site value and habitat quality for ponds, lakes, rivers, and streams (Butler and deMaynadier, 2008; Chovanec et al., 2002; D'Amico

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et al., 2004; Flenner and Sahlén, 2008; Primack et al., 2000; Raebel et al., 2012; Remsburg and Turner, 2009; Rosset et al., 2013; Silva et al., 2010). Bioassessment tools based on adult Odonata have been developed and tested in Europe and South Africa. Chovanec and Waringer (2001) combined species-specific abundance classes, niche width, and habitat preference into an Odonata Habitat Index meant to classify the ecological status of river-floodplain systems in Austria. Simaika and Samways (2009) combined species' geographical range, risk of extinction, and sensitivity to habitat change into a Dragonfly Biotic Index that has been effective for assessing river condition in South Africa (Simaika and Samways, 2011) and the conservation value of ponds and small lakes in Europe and South Africa (Rosset et al., 2013). These approaches show potential for assessing wetland condition, but they have not been tested in that capacity, specifically.

A reliable attribute in the biological assessment of environmental condition is species *conservatism*, referring to the relative sensitivity (vulnerability) of different species to habitat degradation (Cohen et al., 2004; Lopez and Fennessy, 2002; Miller and Wardrop, 2006). Conservatism is commonly associated with floristic quality assessment, wherein a coefficient of conservatism (CoC) ranging from 0 to 10 is assigned to vascular plant species, based on the expert opinion of a team of botanists. High CoC are given to species that are relatively sensitive to habitat degradation, whereas low CoC are assigned to species that are non-native or highly tolerant. The collective conservatism of a species assemblage should, in theory, reflect the ecological condition of a given area (Swink and Wilhelm, 1979; Taft et al., 1997). In the United States, interest in developing and applying CoC for the assessment of wetland condition is rapidly growing (Bried et al., 2012b); yet to date, conservatism has been applied almost exclusively in the context of floristic quality (e.g., Bried et al., 2013; Cohen et al., 2004; Cretini et al., 2012; Ervin et al., 2006; Lopez and Fennessy, 2002; Medley and Scozzafava, 2009; Matthews et al., 2005; Miller and Wardrop, 2006; but see Micacchion, 2004).

In this study we apply the conservatism concept to adult Odonata. We use an extensive opportunistic survey dataset to introduce an objective, empirical method of assigning CoC based on species occurrence and exclusivity to categories of wetland degradation. We then aggregate the CoC into an index of freshwater wetland condition, and evaluate index performance using independent odonate data and metrics of human disturbance.

2. Materials and methods

2.1. Data

We conducted our study in Rhode Island located in the northeastern United States. We relied on data from the Rhode Island Odonata Atlas Project (hereafter “Atlas”) for this study. The Atlas was conducted from 1999 through 2004 as a statewide inventory of adult Odonata administered by the Rhode Island Natural History Survey and the Rhode Island Chapter of The Nature Conservancy (Brown and Briggs, 2013). Professionals and trained volunteers cataloged 135 Odonata species throughout Rhode Island, collecting ~13,000 verified voucher specimens across 1090 aquatic, wetland, and upland sites. As with other citizen-based statewide Odonata inventory projects (e.g., White et al., 2010) or any opportunistic atlas-type surveys (Robertson et al., 2010), sampling effort was not standardized over time or space.

2.2. Generation of CoC and the wetland integrity index

Assignment of CoC using expert judgment relies on specific knowledge of species distributions relative to the degradation of their habitats. Subjectivity and bias are introduced by the

limitations of experience, a focus on geographic or habitat range, perception of habitat degradation, and interpretation of the CoC designations (Bried et al., 2012b). To avoid these problems, we generated Odonata CoC empirically, using georeferenced point records from the Atlas and a Geographic Information System (GIS).

We assigned the CoC based on species' occurrences among freshwater wetlands. To account for dataset spatial inaccuracies and increase the likelihood that sampling points were associated specifically with wetlands, only points that occurred within or near (<50 m) previously mapped wetlands were considered. Points associated with unvegetated surface waters or uplands were excluded from analysis. Qualified points were assumed to be representative wetlands, and were sorted by the proportion of developed and agricultural land within 300 m. Points in the lower quartile were selected as least-disturbed wetlands, points in the upper quartile as most-disturbed wetlands, and an equal number of points surrounding the median as intermediately disturbed wetlands; this resulted in a training sample of 510.

Following the indicator species analysis proposed by Dufrene and Legendre (1997), a CoC was determined for each species by:

$$\frac{(N_{LD}/N) + (1 - N_{MD}/N)}{2} \times 10$$

where N_{LD} is the number of least-disturbed wetlands in which a given species was detected, N_{MD} is the number of most-disturbed wetlands where that species was detected, and N is the total number of wetlands (including intermediately disturbed sites) where that species was detected. This approach averages the “affinity” for least-disturbed wetlands and the inverse affinity for most-disturbed wetlands, multiplying by 10 to scale the output to the traditional CoC scale of floristic quality assessment. Thus the CoC range from 0 if a species occurs exclusively in the most-disturbed group to 10 if a species occurs exclusively in the least-disturbed group. In line with recommendations for floristic quality assessment (e.g., Bried et al., 2013; Rooney and Rogers, 2002; Taft et al., 2006), we recommend the mean CoC of all species found at a particular wetland site as an Odonata Index of Wetland Integrity (OIWI).

2.3. Index performance

To evaluate the OIWI, we used a sample of Atlas wetlands that was independent of the training sample described above. Prior to extracting the training sample, we isolated wetland features that were surveyed at least three times and produced at least 10 specimens over the Atlas project period. From that subset, we selected 51 study sites spanning a gradient of surrounding land use intensity. We used photointerpretation of recent leaf-off, high-resolution aerial imagery to delineate a polygonal wetland assessment unit for each study site according to Kutcher (2011). Wetland assessment units ranged in size from 0.12 to 36 ha with an average of 5.3 ha. Many (43) of the units contained multiple vegetation classes. The most frequently represented vegetation classes (per Cowardin et al., 1979) within the study sample were Emergent Wetland (40 sites), Forested Wetland (37 sites), and Shrub Swamp (36 sites), and the most common hydrogeomorphic settings (modified from Brinson, 1993) were Connected Depression (16 sites), Isolated Depression (16 sites), and Floodplain-riverine (16 sites).

We tested the OIWI against the Rhode Island Rapid Assessment Method, or RIRAM (Kutcher, 2011), which follows federal guidelines for establishing reference conditions for wetlands (Faber-Langendoen et al., 2009; U.S. EPA, 2002). This evidence-based tool produces a relative index of freshwater wetland condition and focuses on estimation, rather than interpretation, to maximize objectivity. RIRAM scoring is based on the premise that diverse human disturbances additively contribute to the degradation of general wetland condition (Fennessy et al., 2007;

Table 1

Components of the Rhode Island Rapid Assessment Method for evaluating freshwater wetland condition.

Metric	Metric scoring criteria
1. Integrity of buffers	Estimates % cultural cover class within 100 ft (30 m) of unit
2. Integrity of surrounding landscape	Generates a weighted average of four land-use-intensity categories by relative proportion within 500 ft (150 m) of unit
3. Impoundment	Estimates water regime change and proportion of unit affected, and identifies barriers to resource movement
4. Draining or diversion of water	Estimates water regime change and proportion of the unit affected
5. Anthropogenic fluvial inputs	Estimates impacts of four types of fluvial inputs including nutrients, sediments and solids, toxins and salts, and flashiness
6. Filling and dumping	Estimates the intensity of fill within or abutting the wetland and the proportion of the unit affected
7. Excavation and substrate disturbances	Estimates the intensity of substrate disturbances within the wetland and the proportion of the unit affected
8. Vegetation and detritus removal	Estimates the extent and the proportion of vegetation and detritus removal from each of five vegetation strata
9. Invasive species within wetland	Estimates the collective cover class of all identified invasive plant species
10. Observed state	Rates the apparent integrity of five wetland functional characteristics, including hydrologic integrity, water and soil quality, habitat structure, vegetation composition, and habitat connectivity

U.S. EPA, 2006). Metrics of buffer integrity (2 metrics), in-wetland stress (7 metrics), and functional integrity (1 metric) are summed to generate a single index based on 100 possible points, with each metric carrying ten points (Table 1). A RIRAM score of 100 indicates no observed stresses or impacts, whereas scores approaching 0 indicate a high degree of degradation, thus RIRAM decreases incrementally with an increase in perceived disturbance. We collected RIRAM data according to Kutcher (2011) at each of the 51 study sites.

Because RIRAM is inherently partly subjective, we also tested the OIWI against the proportion of impervious surface area (ISA) within the surrounding 305 m (1000 ft) of each polygonal wetland unit in the study sample. The relative area of impervious surface provides an effective surrogate for human influence because it summarizes and reflects multiple effects of anthropogenic stress (Karr and Chu, 1997). We generated ISA directly from high-resolution impervious surface data (RIGIS, 2010), resulting in a coarse but objective disturbance measure to support our validation analysis.

2.4. Statistical analysis

Residuals from simple linear modeling of OIWI over RIRAM and ISA for the 51 wetlands showed clear heterogeneity and non-normality based on goodness-of-fit (Shapiro–Wilk test), residual by predicted plots (“cone-shaped” spread), and Q–Q plots (skewed left). For this reason, we used bootstrap resampling to evaluate the linear model fit for the OIWI versus RIRAM and ISA gradients and for the OIWI versus each RIRAM metric individually. We assumed a bootstrap approach would handle the zero-inflation inherent to some of the individual RIRAM metrics. Using Resampling Stats v4.0 (written by S. Blank, ©2012 statistics.com, Resampling Stats Inc., Arlington, VA), the data were sampled with replacement into a new set of cells, shuffling the rows as units. We then fit a simple linear model to this resampled data set and repeated and scored the model fit output (i.e., R^2 or coefficient of determination) for 1000

iterations. We report the 2.5th and 97.5th percentiles of the resampled distribution as a 95% confidence interval for model fit (see also Bried et al., 2013).

The OIWI was further evaluated using box plots of OIWI distributions in relation to RIRAM and ISA reference designations, following Barbour et al. (1996). Reference designations were established using 25th and 75th percentile index values to identify most-disturbed (degraded) and least-disturbed (reference-standard) thresholds, respectively; all other study units were considered intermediately disturbed. The degree of overlap between interquartile ranges and medians of OIWI distributions was used to evaluate OIWI performance. Non-overlapping interquartile ranges within most and least-disturbed designations indicate high sensitivity to disturbance and excellent metric performance, whereas various degrees of interquartile-median overlap indicate lower sensitivity and performance (Barbour et al., 1996; Jacobs et al., 2010; Veselka et al., 2010).

3. Results

3.1. CoC and index values

Odonata CoC ranged from 0 to 10 with a mean $\pm$ SD of 6.4 ± 2.2 (Table 2). Species observed occurrence rates in the 510-site training sample ranged from zero to 23% with a median of about 3%. Only one of the 135 documented Atlas species, *Libellula auripennis*, was not represented in the training sample; this was assigned a CoC of 10, since it was observed only once during the Atlas inventory period at a minimally disturbed site (based on 0% cultural land cover within 300 m). Other rarely represented species were assigned CoC following our methods. OIWI values generated with and without incorporating rarely observed species – i.e., those species with fewer than 20 site occurrences in the Atlas ($n=28$ species), based on a natural break in the data and best professional judgment – were nearly identical (Spearman's rank-correlation test, $r_s=0.99$, $P<0.001$, $n=51$ study sites), suggesting that the inclusion of rare species is unlikely to strongly affect OIWI outcomes. Rare-species CoC were therefore retained in the OIWI to avoid introducing bias or circularity associated with culling rare species according to our best professional judgment or calibration with our disturbance gradients.

OIWI values ranged from 3.74 to 7.15 with a mean of 5.90 ± 0.77 among the 51 study sites (Table 3). Number of species recorded per site ranged from 4 (among 17 specimens collected across four site visits) to 47 (among 124 specimens collected across seven visits). We did not find evidence of association between OIWI values and measures of sampling effort per site, including number of specimens, number of visits, and number of species ($r_s=0.13$ – 0.17 , $P=0.22$ – 0.37). RIRAM scores ranged from 37.9 to 100 with a mean of 79.2 ± 17.0 , and ISA ranged from 0 to 62.4% with a mean of $10.0 \pm 14.0\%$, indicating a broad range of wetland conditions across the study sample. The OIWI, ISA, and RIRAM were each uncorrelated with wetland unit size ($r_s=-0.09$ to 0.04 , $P=0.53$ – 0.90).

3.2. Index performance

The OIWI was well predicted by the overall RIRAM gradient and showed evidence of a linear relationship with the impervious surface area measure (Fig. 1). OIWI also showed evidence of a linear relationship with many of the individual RIRAM metrics, including strong relationships with the buffer, landscape, and integrated functional ('Observed State') metrics (Table 4).

OIWI interquartile ranges within the most-disturbed and least-disturbed wetland categories, as determined by both RIRAM and

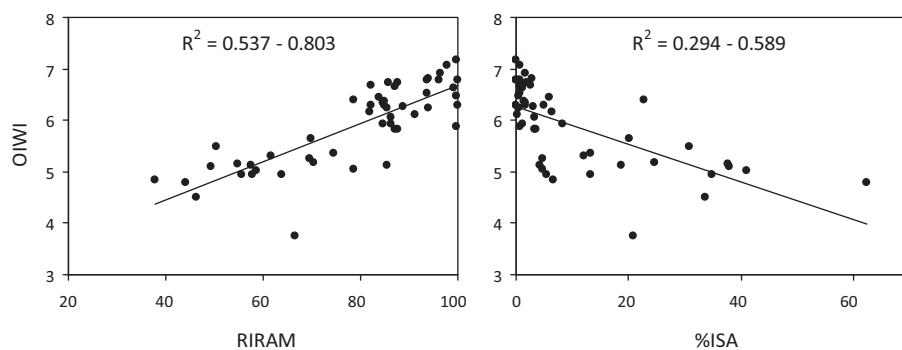
Table 2
Coefficients of conservatism (CoC) for 135 Odonata species known to occur in Rhode Island and the number of training sites where each was collected; LD = least-disturbed, ID = intermediately disturbed, and MD = most-disturbed.

Species	Training sites					Species	Training sites				
	CoC	LD	ID	MD	Total		CoC	LD	ID	MD	Total
<i>Aeshna canadensis</i>	8.3	4	2	0	6	<i>Hagenius brevistylus</i>	7.6	11	7	1	19
<i>Aeshna clepsydra</i>	8.3	16	3	2	21	<i>Helocordulia uhleri</i>	7.7	9	5	1	15
<i>Aeshna constricta</i>	5.0	3	4	3	10	<i>Hetaerina americana</i>	5.0	4	6	4	14
<i>Aeshna mutata</i>	7.5	1	1	0	2	<i>Ischnura hastata</i>	5.4	8	9	6	23
<i>Aeshna tuberculifera</i>	8.2	12	7	0	19	<i>Ischnura kellicotti</i>	5.2	8	8	7	23
<i>Aeshna umbrosa</i>	6.2	11	9	5	25	<i>Ischnura posita</i>	4.1	29	36	51	116
<i>Aeshna verticalis</i>	8.6	14	3	1	18	<i>Ischnura ramburii</i>	0.0	0	0	4	4
<i>Amphiagrion saucium</i>	6.4	6	2	3	11	<i>Ischnura verticalis</i>	3.4	13	35	42	90
<i>Anax junius</i>	5.1	20	21	19	60	<i>Lanthus vernalis</i>	7.5	1	1	0	2
<i>Anax longipes</i>	8.3	5	0	1	6	<i>Lestes congener</i>	5.7	10	6	7	23
<i>Argia apicalis</i>	1.9	0	3	5	8	<i>Lestes disjunctus</i>	6.7	18	15	5	38
<i>Argia fumipennis</i>	4.6	23	31	29	83	<i>Lestes dryas</i>	3.3	0	2	1	3
<i>Argia moesta</i>	2.6	2	6	11	19	<i>Lestes eurinus</i>	8.0	11	2	2	15
<i>Argia translata</i>	2.0	0	2	3	5	<i>Lestes forcipatus</i>	5.9	21	21	11	53
<i>Arigomphus furcifer</i>	6.7	5	2	2	9	<i>Lestes inaequalis</i>	5.8	17	16	10	43
<i>Arigomphus villosipes</i>	5.5	12	9	9	30	<i>Lestes rectangularis</i>	6.2	31	28	14	73
<i>Basiaeschna janata</i>	7.2	18	17	2	37	<i>Lestes unguiculatus</i>	0.0	0	0	2	2
<i>Boyeria vinosa</i>	5.8	9	11	5	25	<i>Lestes vigilax</i>	5.4	28	29	22	79
<i>Calopteryx aequabilis</i>	7.3	6	7	0	13	<i>Leucorrhinia frigida</i>	8.8	15	5	0	20
<i>Calopteryx dimidiata</i>	5.3	7	6	6	19	<i>Leucorrhinia glacialis</i>	10.0	1	0	0	1
<i>Calopteryx maculata</i>	5.7	31	33	20	84	<i>Leucorrhinia hudsonica</i>	7.8	6	2	1	9
<i>Celithemis elisa</i>	5.7	22	18	14	54	<i>Leucorrhinia intacta</i>	6.3	20	19	8	47
<i>Celithemis eponina</i>	4.6	6	9	8	23	<i>Leucorrhinia proxima</i>	8.8	3	1	0	4
<i>Celithemis fasciata</i>	7.7	9	5	1	15	<i>Libellula auripennis</i>	10.0	0	0	0	0
<i>Celithemis martha</i>	6.5	10	2	5	17	<i>Libellula axilena</i>	8.8	3	1	0	4
<i>Chromagrion conditum</i>	6.7	31	21	10	62	<i>Libellula cyanea</i>	6.4	20	15	8	43
<i>Cordulegaster diastatops</i>	8.5	9	4	0	13	<i>Libellula deplanata</i>	8.3	2	1	0	3
<i>Cordulegaster maculata</i>	7.5	7	4	1	12	<i>Libellula exusta</i>	8.1	27	9	3	39
<i>Cordulegaster obliqua</i>	10.0	2	0	0	2	<i>Libellula incesta</i>	5.4	29	28	22	79
<i>Cordulia shurtleffi</i>	8.3	2	1	0	3	<i>Libellula julia</i>	10.0	5	0	0	5
<i>Didymops transversa</i>	7.5	6	6	0	12	<i>Libellula luctuosa</i>	4.0	10	26	22	58
<i>Dorocordulia lepida</i>	8.8	22	5	1	28	<i>Libellula lydia</i>	6.0	26	19	14	59
<i>Dorocordulia libera</i>	10.0	6	0	0	6	<i>Libellula needhami</i>	1.0	0	1	4	5
<i>Dromogomphus spinosus</i>	3.5	3	6	8	17	<i>Libellula pulchella</i>	4.2	7	8	11	26
<i>Enallagma aspersum</i>	5.6	16	14	11	41	<i>Libellula quadrimaculata</i>	8.9	7	2	0	9
<i>Enallagma boreale</i>	7.9	8	3	1	12	<i>Libellula semifasciata</i>	7.5	13	4	3	20
<i>Enallagma civile</i>	4.0	17	23	32	72	<i>Libellula vibrans</i>	5.0	2	3	2	7
<i>Enallagma cyathigerum</i>	7.5	3	3	0	6	<i>Macromia illinoensis</i>	6.0	7	4	4	15
<i>Enallagma daeckii</i>	6.9	8	2	3	13	<i>Nannothemis bella</i>	7.5	6	6	0	12
<i>Enallagma divagans</i>	5.6	20	18	14	52	<i>Nasiaeschna pentacantha</i>	7.1	5	7	0	12
<i>Enallagma doubledayi</i>	5.9	8	3	5	16	<i>Nehalennia gracilis</i>	7.3	21	15	3	39
<i>Enallagma durum</i>	1.3	0	1	3	4	<i>Nehalennia integricollis</i>	10.0	1	0	0	1
<i>Enallagma ebrium</i>	5.7	4	8	2	14	<i>Nehalennia irene</i>	6.0	9	11	4	24
<i>Enallagma exsulans</i>	2.1	1	7	13	21	<i>Neurocordulia obsoleta</i>	7.5	1	1	0	2
<i>Enallagma geminatum</i>	4.7	28	30	33	91	<i>Ophiogomphus aspersus</i>	9.4	7	1	0	8
<i>Enallagma hageni</i>	6.5	5	3	2	10	<i>Ophiogomphus mainensis</i>	8.8	3	1	0	4
<i>Enallagma laterale</i>	6.4	14	8	6	28	<i>Pachydiplax longipennis</i>	4.1	21	22	36	79
<i>Enallagma minusculum</i>	6.1	3	5	1	9	<i>Pantala flavescens</i>	3.1	2	4	7	13
<i>Enallagma pictum</i>	7.5	7	1	2	10	<i>Pantala hymenaea</i>	2.3	0	5	6	11
<i>Enallagma recurvatum</i>	8.2	10	3	1	14	<i>Perithemis tenera</i>	3.9	11	19	23	53
<i>Enallagma signatum</i>	3.7	9	25	24	58	<i>Progomphus obscurus</i>	8.8	3	1	0	4
<i>Enallagma traviatum</i>	4.3	4	12	7	23	<i>Somatochlora georgiana</i>	9.0	4	1	0	5
<i>Enallagma vesperum</i>	4.5	4	9	6	19	<i>Somatochlora linearis</i>	8.8	10	3	0	13
<i>Enallagma weewa</i>	7.1	5	0	2	7	<i>Somatochlora tenebrosa</i>	8.8	24	8	0	32
<i>Epiaeschna heros</i>	6.7	5	2	2	9	<i>Somatochlora walshii</i>	9.0	4	1	0	5
<i>Epithea canis</i>	8.8	3	1	0	4	<i>Somatochlora williamsoni</i>	10.0	3	0	0	3
<i>Epithea cynosura</i>	6.3	32	31	12	75	<i>Stylogomphus albistylus</i>	6.4	8	7	3	18
<i>Epithea princeps</i>	5.8	7	7	4	18	<i>Stylurus scudderii</i>	6.7	1	2	0	3
<i>Epithea spinigera</i>	8.8	3	1	0	4	<i>Stylurus spiniceps</i>	5.0	0	2	0	2
<i>Erythemis simplicicollis</i>	5.3	20	23	16	59	<i>Sympetrum costiferum</i>	4.5	4	2	5	11
<i>Erythrodiplax berenice</i>	3.7	5	7	11	23	<i>Sympetrum internum</i>	5.0	34	34	34	102
<i>Gomphaeschna antilope</i>	7.5	1	1	0	2	<i>Sympetrum rubicundulum</i>	4.2	2	6	4	12
<i>Gomphaeschna furcillata</i>	8.5	16	7	0	23	<i>Sympetrum semicinctum</i>	7.0	13	9	3	25
<i>Gomphus abbreviatus</i>	5.0	1	2	1	4	<i>Sympetrum vicinum</i>	5.6	21	16	15	52
<i>Gomphus adelphus</i>	8.0	3	2	0	5	<i>Tramea carolina</i>	5.3	7	2	6	15
<i>Gomphus exilis</i>	7.1	34	28	6	68	<i>Tramea lacerata</i>	5.0	8	8	8	24
<i>Gomphus lividus</i>	7.8	6	2	1	9	<i>Williamsonia lintneri</i>	7.5	3	3	0	6
<i>Gomphus spicatus</i>	10.0	2	0	0	2						

Table 3

Odonata Index of Wetland Integrity (OIWI) values and effort data from 51 wetland assessment units in Rhode Island; information is listed in decreasing order of OIWI.

Wetland Unit	OIWI	# Visits	# Specimens	# Species	Wetland Unit	OIWI	# Visits	# Specimens	# Species
SMA-ARC-BFFEN	7.16	8	26	17	PRV-BRCH-STA1	6.01	6	64	36
SMA-ARC-WD3	7.06	11	24	14	SMA-ARC-MOON	5.93	7	13	8
AUD-EPP-QR4	6.82	5	11	6	SMA-GWMA-OKPD	5.92	7	32	19
SMA-CAR-WLPD	6.79	9	34	11	SMA-BUCK-PD1	5.88	6	34	21
PRV-BOTH-PND	6.78	7	124	47	TNC-CRTR-WET1	5.83	4	17	4
AUD-FISH-BRK	6.77	5	14	10	AUD-NEW-PND	5.82	4	53	24
TNC-XXX-QR2	6.74	30	69	37	PRV-XXX-PWT5	5.65	6	26	15
PRV-MAIL-FEN	6.72	3	10	5	PRV-SLTR-PRK0	5.49	5	16	11
SMA-ARC-RBPD	6.72	5	62	29	PRV-HUNT-STA3	5.37	5	57	21
PRV-GRSY-PND	6.69	8	19	7	PRV-BUTT-PND	5.32	4	20	12
SMA-BIG-CAP	6.64	18	105	43	PRV-THIR-PND	5.27	4	10	9
TNC-ELL-PND	6.64	3	14	8	PRV-TEN-RIV1	5.17	10	36	19
SMA-DUR-TEPE	6.53	5	55	29	PRV-WOON-STA3	5.14	10	34	16
PRV-PED-PND	6.46	4	28	14	PRV-CARR-PND	5.13	5	19	9
PRV-MOW-BRK2	6.45	5	13	9	PRV-LONS-MRSH	5.13	5	15	10
PRV-HART-BOG	6.40	4	50	24	PRV-EVAN-PND	5.11	4	17	12
SMA-GSW-CHIP7	6.36	3	18	11	PRV-ASHA-RIV2	5.04	6	17	13
PRV-SNAKE-POW	6.34	5	16	8	PRV-XXX-PWT17	5.03	4	15	12
PRV-JACK-SCPD	6.29	3	15	15	PRV-WAR-RES	4.95	14	43	21
SMA-CAR-FISH	6.29	16	37	18	PRV-WOON-STA4	4.95	4	22	11
PRV-R216-POW	6.28	5	16	13	PRV-BLRD-PARK	4.94	8	22	9
PRV-PYSZ-FEN	6.26	10	34	19	PRV-MITC-PND	4.85	3	25	13
AUD-CARD-SWP	6.24	5	41	23	PRV-MOSH-PND	4.78	10	55	17
SMA-WOO-IMP	6.24	17	99	34	PRV-NOTT-PD1	4.50	4	16	11
PRV-GLAC-PND	6.16	8	54	22	PRV-DMCR-PLAY	3.74	4	11	5
PRV-FORG-GRN1	6.10	18	55	23					

**Fig. 1.** Performance of the OIWI: Odonata Index of Wetland Integrity for 51 wetland sites in relation to the Rhode Island Rapid Assessment Method and % impervious surface area (measured in a 305-m buffer around each site); model fit (R^2) is based on computer-intensive resampling with 1000 iterations; best fit line is based on linear regression.

ISA, did not overlap, and median OIWI values differed between those categories according to both indices (Fig. 2; Mann–Whitney U -tests, $Z = -4.33$ and -4.08 , $P < 0.001$). Additionally, the median OIWI in most-disturbed and least-disturbed wetlands differed from the median OIWI in intermediately disturbed wetlands as determined by RIRAM ($Z = 3.49$ and 4.60 , $P < 0.001$). There was no evidence that median OIWI or RIRAM values varied

among connected depression, isolated depression, and floodplain-riverine geomorphic settings (Kruskal–Wallis test, $H = 3.02$, $P = 0.22$ and $H = 1.07$, $P = 0.59$, respectively), indicating that hydrogeomorphology did not strongly bias OIWI or RIRAM outcomes. Vegetation-based classes could not be analyzed in this way because more than one type was often represented within a single study unit.

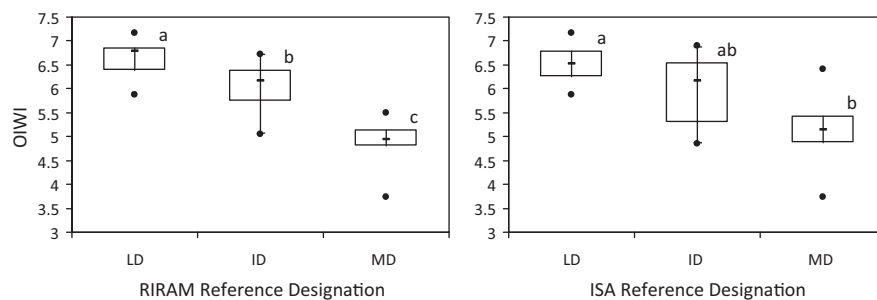
**Fig. 2.** Discriminating among disturbance designations: box and whisker plots depicting the distribution of OIWI values ($n = 51$) in relation to three reference designations derived from RIRAM and ISA values, respectively; LD = least-disturbed, ID = intermediately disturbed, and MD = most-disturbed. The center dash represents the median ($a > b > c$), the box represents the interquartile range, the whiskers represent 2.5th and 97.5th percentiles, and the round symbols represent maximum and minimum values.

Table 4

Confidence limits (2.5th and 97.5th percentiles) of linear model fit between individual RIRAM metrics (see Table 1) and the OIWI based on computer-intensive resampling (1000 iterations); metrics 1, 2, and 10 decreased with increased disturbance, whereas metrics 4 through 9 increased.

Metric	Low R^2	High R^2
1. Integrity of buffers	0.579	0.787
2. Integrity of surrounding landscape	0.507	0.793
3. Impoundment	0.000	0.121
4. Draining or diversion of water	0.128	0.502
5. Anthropogenic fluvial inputs	0.212	0.650
6. Filling and dumping	0.314	0.610
7. Excavation and substrate disturbances	0.013	0.245
8. Vegetation and detritus removal	0.001	0.238
9. Invasive species within wetland	0.183	0.545
10. Observed state	0.539	0.792

4. Discussion

4.1. Index performance

An effective indicator must separate human disturbance and degraded ecological condition from the inherent variation found in nature (Brazner et al., 2007; Karr and Chu, 1999; Swink and Wilhelm, 1979; Taft et al., 1997). Our study demonstrates the potential of a new index (OIWI) for freshwater wetland condition assessment built on the empirically estimated conservatism of adult odonate species. Correlations between OIWI and a multi-metric disturbance gradient (RIRAM) suggest that multiple stressors influence wetland patch use by odonate species. The stronger linear relationship with the full RIRAM than with any of the component metrics suggests the OIWI is more likely to indicate overall wetland condition rather than any particular stressor. And, the clear relationship of the OIWI to the buffer and landscape metrics supports the idea that adult odonates are also strong indicators of land use practices and integrity of the area surrounding the wetland breeding site.

Non-overlapping interquartile ranges suggest excellent capability of the OIWI to discriminate among reference categories, defined according to the RIRAM and ISA measures. Indeed, the entire OIWI distributions within RIRAM-designated least-disturbed and most-disturbed wetlands were non-overlapping. Discriminating among disturbance classes is often a key objective of wetland assessment (Jacobs et al., 2010; U.S. EPA, 2006). The tighter relationship (better model fit) of the OIWI to the RIRAM than to ISA suggests that odonates as a group will respond more predictably to cumulative in-wetland and adjacent (<150 m) stresses than to broader (300 m) surrounding landscape stresses, even though the CoC were generated at the latter scale. This supports the fact that much adult odonate activity and abundance is localized in and around breeding habitat (Bried and Ervin, 2006; Butler and deMaynadier, 2008), and undermines the prevailing opinion that adult stages cannot indicate conditions at the breeding site (Raebel et al., 2010). Strong correlations between OIWI and RIRAM buffer metrics suggest that adult Odonata are highly sensitive to the condition of nearby uplands surrounding the breeding site. This contrasts with odonate larvae which may respond only or primarily to breeding site conditions (Raebel et al., 2012). We recommend a full evaluation of adults vs. larvae (or exuviae) based on concurrent sampling of both stages along the same disturbance gradient.

Simaika and Samways (2011) found that adult dragonfly species composition, as represented by the Dragonfly Biotic Index, was more efficient and effective than benthic macroinvertebrate composition for assessing river condition. Similar to the OIWI, their index incorporates aggregate sensitivity of adult odonates to human disturbances. Metrics evaluating geographical range and threat of extinction, typically associated with habitat conservation

value, collectively outweigh the species sensitivity metric. But, it is unclear how these metrics affect the signal of human disturbance, because they may correspond with conservatism, in that conservative species may be restricted in geographical range or threatened, due to habitat degradation. The Odonata Habitat Index (Chovanec and Waringer, 2001), intended to assess the health of river-floodplain systems, incorporates metrics evaluating species abundance, niche width, and habitat preference. While niche width may correspond with conservatism, species abundance and habitat preferences are heavily weighted, shifting the index focus toward habitat suitability for Odonata and away from general ecological condition. In contrast to these methods, the OIWI uses only collective species sensitivity as the indicator, thereby inherently restricting the index assessment to site quality. Accordingly, any effective use of a wetland by adult Odonata was counted in generating the CoC and validating the OIWI. Although the OIWI performed well without separating resident (autochthonous, successfully emerged) and immigrant species, a validation analysis focused strictly on the resident assemblage may find an even better signal of site quality. This is because the in-wetland stress experienced during the larval period may carry over to determine the species present at the adult stage.

Our study indicates the potential value of adult Odonata species conservatism as an effective and efficient indicator of freshwater wetland condition. We propose that the OIWI may provide a reliable alternative or complement to the conservatism-based floristic quality indices that have become popular for wetland assessments in the United States (Bried et al., 2013; Cohen et al., 2004; Ervin et al., 2006; Lopez and Fennessy, 2002; Miller and Wardrop, 2006; Stein et al., 2009). The linear model fit between OIWI and measures of human disturbance was comparable to wetland assessments using floristic conservatism (e.g., Cohen et al., 2004; Ervin et al., 2006; Lopez and Fennessy, 2002; Miller and Wardrop, 2006). Because adult odonates require the habitat surrounding wetlands for maturation, foraging, nocturnal roosting, and other activities (Bried and Ervin, 2006 and references therein), and because the CoC are estimated objectively rather than using best professional judgment, the OIWI may provide a more integrated and accurate measure of wetland quality than site-restricted floristic assessments. A direct comparison of the OIWI and floristic quality methods is needed to test this prediction. Furthermore, the OIWI uses a readily observed insect group whose species identifications are easy to learn, and thus should not present any greater logistical difficulty than floristic-based assessments. However, we acknowledge that odonatists are outnumbered by botanists, and that odonates may not be present in all types of wetlands.

Similar to some floristic methods, OIWI is a straightforward, single-metric indicator of wetland condition that is easily understood and thus may be a more intuitive tool for practitioners than more complex indicators. The OIWI is based on the straightforward premise that because Odonata species exhibit differential tolerance to various human disturbances, species assemblage can reflect cumulative human disturbance at a given wetland. Assignment of CoC was also straightforward, based on the empirical analysis of species occurrences using observational data. Bioindicators that employ numerous metrics, complex metrics, or metrics based on a coarse or subjective characterization of condition (such as expert opinion) are more likely to contain biases and hidden information that cannot easily be understood and reconciled by the end user. Practitioners may therefore feel more confident applying the OIWI over more complex or subjective indicators.

4.2. Methodology considerations

We used the mean CoC for the OIWI and ignored species richness, which for odonates may correspond with site attributes other

than ecological condition (Aliberti-Lubertazzi and Ginsberg, 2010; Bried et al., 2007; Hornung and Rice, 2003; Sahlén and Ekstubbé, 2001). For example, several odonate studies have reported a positive relationship between number of species and patch area (Bried et al., 2012a; Kadoya et al., 2004; Samways et al., 2011). Additionally, the number of adult odonate species observed depends largely on the frequency and duration of surveys (Bried et al., 2012a; Simaika and Samways, 2009). Survey effort and assessment unit size varied greatly in the Rhode Island Odonata Atlas, but neither correlated with the OIWI, suggesting that these discrepancies did not affect OIWI values relative to our disturbance gradients; however, we hypothesize that patch area and sampling effort variability would confound the index if it incorporated species richness. Studies of floristic quality have also recognized the confounding influence of richness and recommended using mean CoC alone (Bried et al., 2013; Cohen et al., 2004; Miller and Wardrop, 2006; Rooney and Rogers, 2002).

A main goal of our study was to develop accurate Odonata CoC for practical application in wetland assessment. We therefore used three training groups, representing least-disturbed, intermediately disturbed, and most-disturbed wetlands, to maximize CoC information under the data constraints of the Odonata Atlas. However, in applications collecting new Odonata training data or utilizing a more rigorous survey dataset, it may be more efficient and effective to use only least-disturbed and most-disturbed groups, at the expense of losing information from intermediately disturbed wetlands. Advantages could include a reduction in ecological noise, more efficient, targeted monitoring effort, and simpler CoC computations, using a single proportional value of affinity rather than averaging two (affinity to least-disturbed wetlands would automatically correspond to inverse affinity to most-disturbed wetlands).

Our method of empirically assigning CoC could be applied to other large opportunistic or “citizen-science” datasets for Odonata, or to similar datasets for other species assemblages. For example, Micacchion (2004) used best professional judgment to assign coarse CoC to amphibian species to indicate the condition of seasonally flooded ponds in Ohio, USA. Many states, including Ohio, have extensive spatially referenced amphibian datasets that could be utilized for assessment by applying our methods to generate amphibian CoC. Similarly, Lussier et al. (2006) assigned subjective coefficients of tolerance to songbird guilds to help describe the ecological integrity of riparian corridors. Our methods could be applied to the extensive, existing songbird datasets to empirically assign CoC to individual bird species, which could potentially facilitate rapid assessment of large conservation areas using analysis of existing spatial data or new songbird point-counts. Also, floristic CoC could be validated or improved using similar methods (Bried et al., 2012b), although this could be an onerous task that would need to be weighed against potential benefits over expert-based CoC. Cohen et al. (2004) found negligible functional differences between index values using data-based versus opinion-based CoC for plants.

There are expected disadvantages to using odonate adults relative to larvae and exuviae. Flight activity is sensitive to weather conditions and may affect species' detection probability, generating noise in the data set. Also, presence of adults or their mating and oviposition attempts do not indicate successfully emerged or breeding resident species (Chovanec and Waringer, 2001; Raebel et al., 2010). Separating the resident and immigrant species may improve OIWI performance, but currently no criteria exist for doing so based only on adult surveys.

It is unclear whether species with low representation in the training sample were given accurate CoC. Although the likelihood of any one or combination of these species strongly affecting OIWI outcomes across multiple wetlands is low, rare species may provide vital information for site-scale assessment (Poos and Jackson,

2012). Incorporating rare species allowed us to test the application of all available species information, which may be important for assessing wetlands with low species richness. Similarly, Simaika and Samways (2009) found that the Dragonfly Biotic Index was not substantially affected by occasional species, even as rarity (in terms of relative geographic distribution and conservation status) is heavily positively weighted in the index. In contrast, our empirical method of CoC allocation will favor rare species over common species only if they are primarily observed in undisturbed landscapes.

The number of species documented at certain study units may be biased low due to targeted sampling of early season species during the Atlas (V. Brown, pers. comm.). In fact, the observed number of odonate species is likely biased low at any sites with one or few surveys. But if we assume this bias is evenly distributed (approximately) across the sample, then our novel approach to CoC designation can be applied using many large opportunistic data sets that already exist (e.g., White et al., 2010). A standardized sampling effort for adult Odonata over the flight season (see Bried et al., 2012a for guidance) at an independent set of wetlands could then be used to rigorously evaluate the performance of CoC estimated from opportunistic data.

4.3. Conclusion

Many forms of wetland bioassessment exist with varying levels of complexity and required expertise (Rader et al., 2001; U.S. EPA, 2002). Our study demonstrates a straightforward and effective method of empirically assigning CoC to odonate species based on their affinity to disturbance classes assigned to a large opportunistic dataset. We found that adult Odonata sensitivity to disturbance, taken collectively across species, responds predictably to multiple aspects of wetland and adjacent buffer degradation, and declines monotonically in response to cumulative wetland degradation (i.e., general wetland condition) across a range of freshwater wetland types. These findings indicate the utility of adult Odonata as a meaningful and robust indicator of freshwater wetland condition. In addition to developing the CoC and testing the OIWI in other regions, future studies should compare the OIWI with the related floristic quality indices (Ervin et al., 2006; Taft et al., 2006), and with multi-metric or multi-taxa indices (e.g., Brazner et al., 2007; Johnston et al., 2009) to evaluate how wetland assessments involving only adult odonates perform in relation to approaches requiring more taxa and expertise.

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